

FSL COMMAND CENTER

Advisor Evidence Portfolio

PATENT PENDING — U.S. PROVISIONAL APPLICATION NO. 64/063,037

Decentralized infrastructure for sovereign data governance — using regulated behavioral health as a deliberate proving ground.

Sovereign by Design

Entity	Future Systems Lab, LLC (Wyoming)
EIN	42-2050630
NPI	1497696264 (Taxonomy 175F00000X: Naturopath)
Principal	Dr. Meg Montañez-Davenport, D.N.Psy., BCHN, CBHP
Patent	64/063,037 — filed May 11, 2026 (non-provisional deadline Oct 29, 2026)
Trademark (42)	USPTO Serial 99533250 — Class 42, Software/Tech, publishing May 19, 2026
Trademark (35)	USPTO Serial 99821948 — Class 35, Marketplace, filed May 13, 2026
GitHub	github.com/Future-Systems-Lab
Contact	future.systems.lab@proton.me

TARGET ADVISORY COMMITTEE

Ira A. Fulton Schools of Engineering — Arizona State University

Dr. Dragan Bošković

Blockchain Research Lab, ASU SCAI

Lens: Decentralized Architecture

Convergent architecture via different cryptographic primitives — public Ethereum + EIP-191 vs. permissioned Hyperledger Fabric + proxy re-encryption. 9 deployed contracts, 17-agent autonomous council, earned-only token model, BenevolenceFund 44/33/22/1 distribution.

Dr. Gail-Joon Ahn

Center for Cybersecurity & Trusted Foundations, ASU SCAI

Lens: Zero-Trust Consent Architecture

EIP-191 wallet-signed consent as unified auth/consent/session/attribution mechanism. Zero-custodial design. Phase 5 bilateral authentication is the doctoral contribution — genuinely unsolved in decentralized identity. Documented threat model with pre-mainnet transparency.

Dr. Hassan Ghasemzadeh

EMIL Lab, ASU College of Health Solutions

Lens: Behavioral Health Data Architecture

EncryptHealth participant-sovereign records, HypnoNeuro L1/L2/L3 framework keyed to neurotransmitter systems + ECS baseline, NeuroBalance biosensor consent scaffold. ML collaboration surface: pattern detection on anonymized consented behavioral data.

9 DEPLOYED SMART CONTRACTS — ETHEREUM SEPOLIA

All contracts independently verifiable at eth-sepolia.blockscout.com

#	Contract	Address	Purpose	Status
1	HypnoNeuroToken (HNT)	0x1ae1...84e2	ERC-20 wellness token	Deployed
2	EHTv2	0x9358...0d88	ERC-20 platform token (144K cap)	Deployed
3	MindMasteryNFT	0xCb9E...7771	ERC-1155 achievement credentials	Deployed
4	SovereignLedger v2	0x4afA...aCc4	Session governance + attestation	Deployed
5	AlchemistForge	0xE092...A324	Permissionless shadow integration	Deployed
6	BenevolenceFund v2	0x96E8...251B	3% community wellness treasury	Deployed
7	SovereignAchievement	0xC3F1...9B8D	ERC-1155 soulbound (20 tiers)	Deployed
8	NeuroBalanceConsent	0x2157...96b8	Biosensor consent scaffold	Scaffolded
9	SovereignSession	0xbEb1...65A1	Guide-initiated session attestation	Deployed

PHASE STATUS — per deep-dive 2026-05-14

Phase	Label	Status
1	Contract Deployment	DEPLOYED
2	Bilateral Consent Capture	SCAFFOLDED
3	EncryptHealth Integration	SPECIFIED
4	Session Attestation	SPECIFIED
5	Dual Multi-Sig + Threshold Encryption	OPEN RESEARCH PROBLEM

PHASE 5 — DOCTORAL RESEARCH CONTRIBUTION

Unsolved Research Problem — Primary Doctoral Contribution

Both Sovereign Guide AND Participant produce EIP-191 signatures over the same session payload (bilateral, temporally bound, written on-chain). Session recording is client-encrypted; the encryption keys are threshold-derived from both signatures, so neither party — and not FSL — can decrypt the recording unilaterally.

Why this is an open research problem:

- EIP-191 is single-signer by design — no canonical pattern for mutual signing bound to a session.
- EIP-1271 validates contract-account signatures — wrong shape for two independent EOA signatures co-bound to a session payload.
- Existing multi-sig wallets (Safe, etc.) solve treasury, not session attestation. Threshold signature schemes (FROST, GG18/GG20) exist as cryptographic primitives but no deployed consent pattern combines them with EIP-191 mutual session auth and client-encrypted recording without centralized key custody.
- The intersection — wallet-signed mutual session auth bound to client-encrypted media, no key escrow — is the genuine gap.

Status: Phase 1 deployed and operational on Sepolia (SovereignSession contract verified, deep-dive 2026-05-14 confirms Phase 5 NOT IMPLEMENTED). Phases 2–4 scaffolded/specified. Phase 5 is the unsolved layer and the applied-project contribution submitted with the ASU DEng (Spring 2027) application.

18-MONTH HORIZON

Target	Item	Advisor Alignment
Jul 2026	BS MIS conferral (CSU Global, GPA 3.94)	All three
Jun-Jul 2026	BHTY paper journal submission	Ghasemzadeh
Oct 1, 2026	ASU DEng application submission	All three
Oct 2026	Phase 5 bilateral auth specification	Ahn (primary), Boškovi█
Oct 2026	Non-provisional patent filing	Ahn
Q4 2026	Mainnet migration (security audit required)	Boškovi█
Spring 2027	ASU DEng enrollment	All three
Q2 2027	First external Sovereign Guides	Ghasemzadeh
Q2 2027	NeuroBalance wearable integration	Ghasemzadeh, Ahn
2028+	Domain-agnostic licensing framework	Boškovi█
2028+	IRB-approved behavioral health study	Ghasemzadeh

AI USE DISCLOSURE

```
// SPDX-License-Identifier: Sovereign
pragma sovereignty ^1.0.0;

contract AIDisclosure {
    address public author = MegMontanezDavenport;
    string[] public tools = ["Claude (Anthropic)"];
    string public intent = "Drafting assistance, structure refinement";

    function ownership() public pure returns (string memory) {
        return "Ideas, lived experience, and framing: 100% human.";
    }
}
```

All architectural decisions, factual claims, and final manuscript content are the responsibility of the author.